COIT12209

LECTURE NOTES

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1 What is Data Science?

1.1 Data Science:

The collection, preperation, analysis, visualization, management and presentation of large collections of data. Data science specifically is the exploration of data via the scientific method. It is crucial we understnad data science as we are in an age where everything is going digital. For example, targeted advertising and sponsored results.

1.2 History of Data Science:

FAANGY (Facebook, Apple, Amazon, Netflix, Google and Yahoo!) realised they have copious amount of data laying around. If they applied machine learning techniques to the data, significant business benefits occur. These can include better product recommendations, fraudulant activity monitoring, and click through rates.

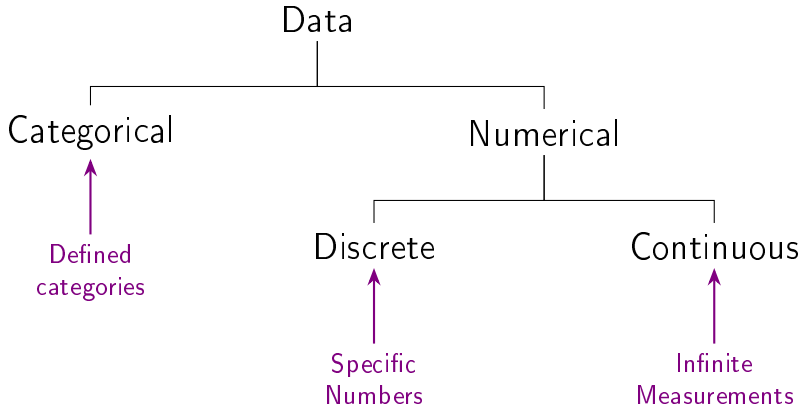
1.3 Data Science Sub-Fields:

- Mathematics and Statistics
- Machine Learning
- Software Engineering
- Domain Knowledge

1.4 What is Data?:

Data usually refers to raw/unprocessed facts. Once anything is done to these facts, in way normies can understand, these facts are considered information.

1.4.1 Data Types



1.5 Big Data:

Comprised of 3 main characteristics:

1. Volume
2. Velocity
3. Variety

Structured Big Data is the common. The formats this comes in is SQL, or CSVs. The positives of these formats is that they can be easily searched. However, must follow strict schema, offering limited flexibility.
Semi-structured have a litt bit of organization like tags or keys, but does not follow strict schema rules. This giver more flexibility too the data but makes it harder to query. Formats can include XML, YAML, and JSON.
Unstructured data follows no consistent formatting. These include text files, PDFs and A/V files. They are rich in formation but hard to store and query efficiently.
Big Data plays a crucial role in industry as it allows data driven descion making.

2 Identifying Data

2.1 Data Mining Task:

Data mining tasks are specific operations or processes to find extra patterns or insights from the data. By breaking down big data into smaller chunks an easier

insight can be found. This allows humans to only do the part where creativity can be found, I.E engaging with stakeholders.

2.2 Business Problems:

Business analystics and problem solving require data scientists to find "correlations" between variables, their meanings and relationships to other variables. In additon it allows informed descion making based on a data driven approach.

2.3 Data Mining Task:

Data mining taks are designed to extract insights or knowlegde from a data set. This is split into 2 categories:

1. Predictive tasks: Use historic data to predict future values
2. Descriptive tasks: Discovering patterns without prior labels

2.3.1 Predictive Tasks

- Classification and class probability: Predicting a category label for each instance.
- Regression: Predict a continuous numeric value.
- Similarity matching: Predict which past records are most alike a new input.
- Link prediction: Predicting future or missing connections in a network.
- Casual modeling: Estimation of the effect of an invention.

2.3.2 Descriptive Tasks

- Clustering: Discovering groupings or clustering within data sets.
- Co-occurence grouping: Items found frequently together.
- Profiling (Behaviour description): Describes typical behaviour within a group.
- Data reduction: Simplify or compress data while preserving key information.

2.4 Classification:

Predicting discrete class labels from the class probability. That being the likelihood that an instance belongs to a particular class.

2.5 Data Collection:

1. Manual Collection
    - Observing and recording
    - Interviews
    - Questionares
    - Pros:
        - Fine-tuned
        - Accurate
    - Cons:
        - Scale
        - Cost
        - Reachablility
2. Automatic Approach
    - Transaction history or logs
    - Sensors suchas, GPS, IoT or Lab equipment
    - Web crawler
    - Social media
    - Pros:
        - Mass collection
        - Easy
        - Cheap on a large scale
    - Cons:
        - Mass collection comes with increased mess

2.6 Sampling:

Data sampling is a statistical analysis technique used to select, manipulate and analyse a subset of representative data. This is done to reduce compute time, resource allocation, and time limits. Sampling types include:

- Simple random sampling
- Stratified sampling
- Cluster sampling
- Systematic sampling
- Multi-stage sampling

2.7 Data Mining Process:

Industry standard is CRIP-DM. This stands for Cross Industry Standard Process for Data Mining.

3 Ethics and Law

3.1 Ethics, Morality and Law:

Ethics is purely a branch of philosophy that consist of a set of values to define the rightness and wrongness of an action. Morality is th ethics and belief held by a group, and usually constituated from cultural and relgious principles. Law is defined by a set of rules determined by a governing body. Usually informed through legal instituation and carries consequence to those who break it.

3.2 Organizational Evil:

OE involves systemic or institutional wrong doing that harms consumers. Basically designed to maximise a companies profits. OE can take various forms:

- Unethical practices
- Abusive behaviour
- Corruption
- Disegard to human-rights.

3.3 Digital Era Bias:

Bias is shown through favouritism toward opinions, action or descions.

1. Algorithmic Bias: AI powered systems ignore bias training data to favour particular outcomes.
2. Sampling Bias: Showed dataset containing societal inequalities. Usually happens because of wrong sampling processes.
3. Confirmation Bias: Recommendations enforcing existing beliefs.
4. Representation Bias: Occurs when certain groups or characteristics are over represented.
5. Accessibility Bias: Design choices affecting those with a disability.
6. Language Bias: LLMs may exhibt a bias towards a particular language.

3.4 Gen AI Ethics:

Gen AI should address concerns related to its generated content.

3.5 Ethical Dilemma:

This occurs when tow or more ethical principles conflict. A solution can be difficult to find.

3.6 Ethical Data Science:

EDS refers to a set of moral principles and guidelines outlining data use. Its about making companies think about utilizing data fairly. Australia subscribes to the follow 10 rules for data research:

1. Acknowledge that data are people and can do harm.
2. Recognize that privacy is more than a binary value.
3. Guard against the reidentification of your data.
4. Practice ethical data sharing.
5. Consider the strengths and limitations of your data; big data does not automatically mean better.
6. Debate the tough, ethical choices.

7. Develop a Code of Conduct for your organization, research community, or industry.
8. Design your data and systems for auditability.
9. Engage with the broader consequences of data and analysis practices.
10. Know when to break these rules.

4 Data Presentation

4.1 Data and Computers:

Computers deal with many different types of data when processing commands and presenting answers. These can be text, audio, video or pictures.

4.2 Analog and Digital:

Information is presented in 2 main ways:

1. Analog: A continuous representation that changes smoothly over time.
2. Digital: Discrete values, that are broken up using binary code.

4.3 Representing Data:

4.3.1 Text

The most common is the ASCII set. Since its introduction in 1968 it has become the default way of doing text with 1Byte per character.

4.3.2 Audio

Soundwaves are an analog signal. Computers need to convert the analog signal to digital for storage and back again for use. It does this through taking small discrete samples at regular intervals.

4.3.3 Video

A video file consists of multiple sub files. Suchas, photos, audio, and subtitles.

4.4 Effictive Presentation:

Raw and processed is not interesting or easy to understand for normies. Making this data easy to digest and understand is critical. Effective data presentation follows the following principles; clear, concise, attractive and attentionable.

5 Data Analytics

5.1 What is it?:

Data analytics is simply the study of data to answer business questions. Not just about more data but a look at the same data. It can be simply put in 5 key principles:

1. Define the analytical problem clearly
2. Collaborate with quantative experts
3. Recognize data types and their impact
4. Apply the right type of analytics
5. Leverage analytics internally and externally

5.2 Exploratory Data Analytics:

This is about getting an overall sense of the dataset. Some of the ways this is done is through visualization and 5-number summaries. This is especially useful in early stages of data mining, finding outliers, missing values and testing assumptions. EDA measures 2 main things:

1. Central tendency: Report a single dat point that describes the most typical response
2. Variability: The typical difference between data points

5.3 Descriptive Modelling:

It looks towards the "what happened" to cause the data trends.

5.3.1 Association Rules

- If/then statements to uncover relationships between data points
- Data sources
- Results: Dependency rules which predict occurrence of some items based on others

5.3.2 Clustering Analysis

- Group data points together that
  - Are more similar to one another
  - In seperate clusters not alike each other
- Similarity Measurements
  - Euclidean distance
  - Problem-specific measurements

5.4 Diagnostic Analysis:

The process behind understanding the causes and reasons behind why a certain event happens. Diagnostic and descriptive analysis work together to find the answer.

5.5 Predictive:

Based on previous history and performance, how is the likelihood going to occur.

5.6 Perscriptive:

Looks forwards for a best course of action based on data analysis.

5.7 Approach to Enterprise Analytics:

- Define business objectives
- Data strategy and governance
- Infrastructure and tools

- Data integration and centralization
- Advanced analytics
- Data visualization and reporting
- Cross-functional collaboration
- Continuous learning
- Feedback and iteration

6 Data Discovery and Mining

6.1 Data Driven Descions:

The process a business makes when taking critical descions, based on reliable data, insights and facts. A descion-making approach that collect analyses and interprets data around KPIs.

6.2 Data Discovery:

The real starting point, starts with a question before data is associated.

6.3 Data Selection:

- Identify relevant data sources for the problem
- Choose attributes that are meaningful
- Ensure data is representative of domain

6.4 Data Cleaning:

- Handle missing values
- Detect inconsistencies and errors
- Remove duplicates
- Address noise, filler and outliers
- Standardise formats

6.5 Data Transformation:

- Feature creation: Generate new attributes to capture useful patterns
- Data reduction: Simplify data sets using dimensionality reduction
- Data normalisation and scaling: Adjust values to a common scale for improved comparability
- Data encoding: Convert categorical into numerical values
- Aggregation/Discretion: Combine on group values

6.6 Data Mining:

A core step of KDD: apply computational techniques to extract patterns, trends or models to data. These methods include:

- Classification: Predict categories
- Regression: Predict continuous values
- Clustering: Group similar records
- Association Rules: Find co-occurance patterns
- Anomaly Detection: Identify unusual cases

6.7 Interpretation and Evaluation:

- Interpret results
- Evaluate usefulness
- Validate models
- Compare alternatives
- Communicate findings

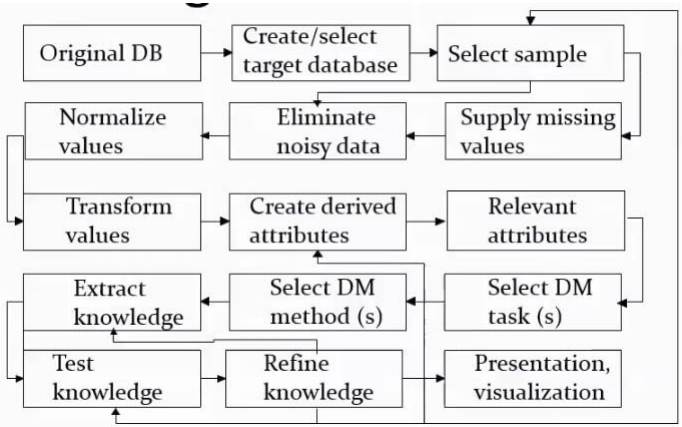
6.8 Data Mining Approach:

A verification driven process:

- Validates a hypothesis
- Querying and reporting
- Multi-dimensional analysis
- Statistical Analysis

It focuses on data-driven discoveries:

- EDA
- Predictive Modelling
- DB segmentation
- Link analysis
- Deviation detection



6.9 Why Data Mining?:

Human analysis breaks down with increased volumn and dimensionality.
Works to be done by non-staticians.

6.10 Data Mining Techniques:

- Association Rules: Detects attributes frequenty found together
- Sequence Mining: Discovers event sequences that occur together
- Similarity Search: Given a DB of objects and a query. Find all occurences within a user-defined range
- Deviation Detection: Find records that are most different from the rest
- Classification and Regression: Assign a new data record to one of several predefined categories or classes
- Clustering: Partition the dataset into subgroups that have common sets of properties

6.11 Data Mining Models:

- Descion Trees
- Random Forrest
- Nueral Network
- K-means Clustering
- K-nearest neighbours
- Apriori and FP-Growth
- Sequence Mining Techniques

6.12 Data Mining Types:

General descriptive knowledge:

- Summarization
- Symbolic descriptions of subsets

Discriminative knowledge:

- Distinguish K-classes
- Accurate classification
- Seperate space

6.13 Components of Data Mining Methods:

- Representation: Language for patterns or models
- Evaluation: Scoring methods for deciding what is a good of model to data
- Search: Method for enumerating patterns or models

7 Exploratory Data Analytics

7.1 Analysis:

The process of looking at data and showing it in tables, charts and simple descriptions. This is so that it become useful information. For example, taking raw exams scores and presenting them in a table.

7.2 Interpretation:

This is understanding what that useful information actually means. Usually done by making connections, comparing results and thinking about reasonable outcomes. Suchas, noticing students who study more achieve better grades and discussing why this might be the case.

7.3 How to Interpret Data?:

- Target: Does result meet a set goal?
- Relevance: Why is this finding relevant?
- Reason: What might explain this result?
- Comparison: How does this result compare with other known good results?
- Triangulation: Are there any other datasets that could be checked to confirm this result?
- Next Steps: Is deeper analysis required to understand the data?

7.4 Trend Analysis:

A basic technique used to patterns, trends and outliers in time series data. Insights flagged will be then placed under deeper investigation.

7.5 Box Plots:

They quickly visualize variations in base data. In addition to quickly identifying outliers. Graphically showing groups of numerical data through their quartiles.

7.6 Spatial Analysis:

Uses location based data to analyse activities and conditions. For example: Map data

- Roads
- Building
- Environment
- Weather

7.7 Pairs Plot:

Helps spot potential correlations by comparing variables pair by pair. Uses non-leading diagonal (wher some variables meet) as a historgram or KDE plot. All others are scatter plots with trend lines.

7.8 Time Series:

Time series is a series of data points in which each data point is associated with a timestamp.

7.8.1 Trends

- Long-term movement of data
- Changes of direction

7.8.2 Seasonal

- Regular repeating patterns tied to a calender
- Frequency is fixed and predictable

7.8.3 Cyclic

- Longer term rises and falls that are not fixed length
- Often linked to business and economic conditions

8 Analytics Algorithms

An analytical model is statistical model that is designed to perofmr a specific task, or predict the probability of an event. Type include:

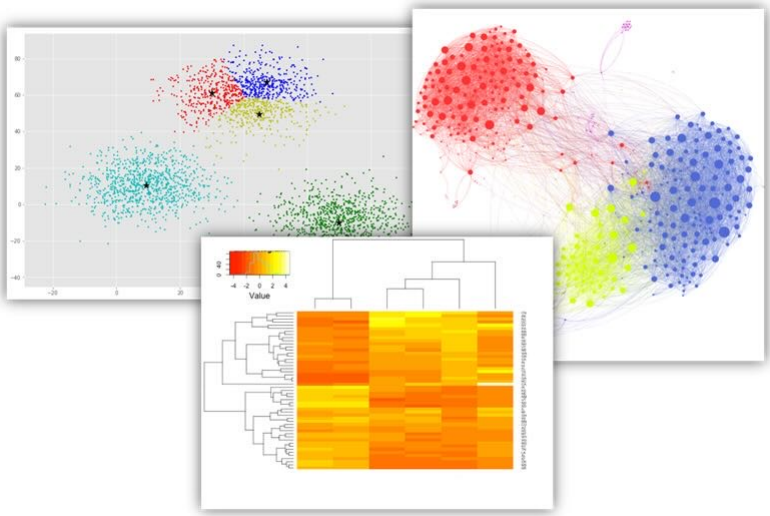
- Classification Model
- K-means clustering Model
- Forecast Model
- Nueral Networks
- Decision Trees
- Logistic Regression
- Time-series Regression

8.1 Clustering Analysis:

CA is the process of grouping a set of objects go that;

- Objects wihin the same group are more similar to each other
- Objects in different groups are less similar

The purpose of CA is the uncover the natural grouping within the data.



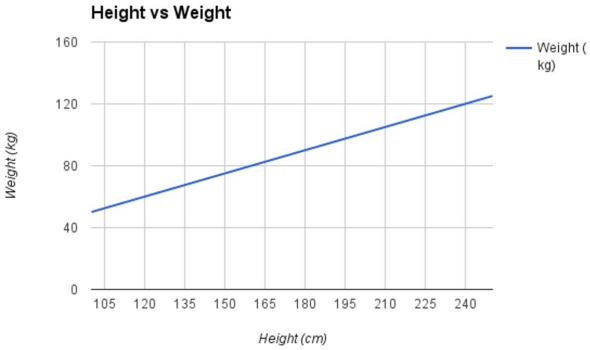
8.2 K-Means Clustering:

One the simplest clustering algorithms.

1. Selects k-number of objects from the dataset, each representing intial cluster centers.
2. Each object is assigned to the cluster to which is most similar. This is based on distance between object and cluster center.
3. The means of clusters are computed as new cluster centers.
4. Function continues until criterion function converges.

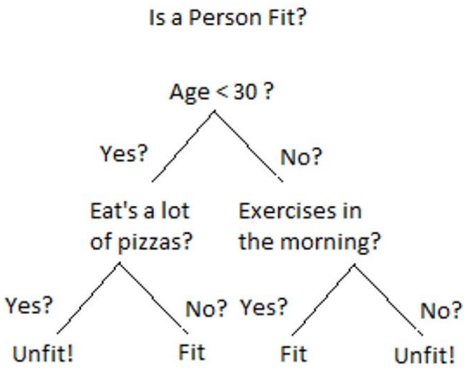
8.3 Regression Analysis:

Used to predict a continuous value such as the volume of rain. Regression is to build a function of independent variables (predictors) to predict a dependent variable (response).



8.4 Classifier Analysis:

Descision classifier uses a descion tree as a predictive model that maps observations about an item to conclusions about the item's target value.

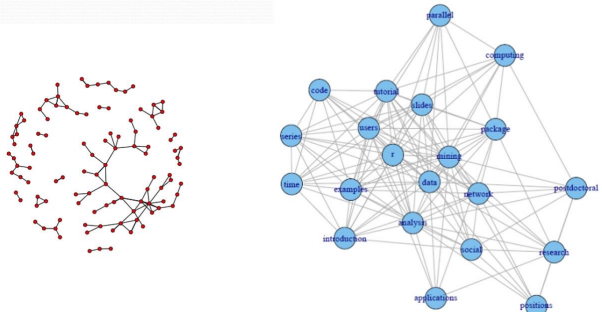


8.5 Association Analysis:

A popular algorithm for discovering a quatifying relationships between variables in large DBs. It shows events or activities that tend to happen together. E.g. the market basket analysis.

8.6 Graph Analysis:

GA uses mathematical structures to model pairwise relations between object. A "graph" in this context is made up of "vertices" or "nodes" and line called edges to connect them.



8.7 Cohorts Analysis:

CohA is the study how cohorts that share common characteristics.

Cohort	Users	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Total Users who did a Transaction	1400	100%	43%	13%	20%	25%	0%	0%	0%
Jan 31	50	100%	0%	0%	0%	0%	0%	0%	0%
Feb 01	100	100%	100%	0%	100%	100%	0%	0%	
Feb 02	200	100%	50%	0%	0.00%	0%	0%		
Feb 03	400	100%	25%	25%	0.00%	0%			
Feb 04	200	100%	100%	50%	0%				
Feb 05	400	100%	25%	0%					
Feb 06	50	100%	0%						

Cohort	Users	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Total Users who abondoned a cart	1110	100.00%	3.98%	1.71%	0.00%	3.00%	1.75%	2.82%	0.00%
Jan 31	100	100.00%	0%	10.26%	0.00%	0.00%	5.26%	5.63%	0.00%
Feb 01	110	100.00%	5.26%	0.00%	0.00%	12.00%	0.00%	0.00%	
Feb 02	190	100.00%	7.60%	0.00%	0.00%	0.00%	0.00%		
Feb 03	130	100.00%	25%	0.00%	0.00%	0.00%			
Feb 04	190	100.00%	100%	0.00%	0.00%				
Feb 05	190	100.00%	15.00%	0.00%					
Feb 06	200	100.00%	0.00%						

Behavioral Cohorts: Customer Retention Analysis

8.8 Traverse Pattern Analysis:

TPA is an example of combining multiple analytic algorithms to betten understand patterns.

Traverse analysis links a customer or product usage patterns and association rules to a geographical or facility map in order to identify potential purchase, traffic, flow, fraud, theft, and other usage patterns and relationships.

The process starts by creating association rules from the customer's or product's usage data, and then maps those association rules to a geographical map (store, hospital, school, campus, sports arena, casino, airport) to identify potential performance, usage, staffing, inventory, logistics, traffic flow, etc. problems.

9 Data Integrations

No particularily related to EDA and ML but still a crucial part of the learning process.

9.1 Objectives:

- Why do we integrate data?: Understand business and technical drivers for combining multiple data sources.
- Challenges to Integration: Issues suchas quality, heterogenity and governences.
- ETL Process: Break down extract, transform and load phases in their role of data preperation.
- Support Concepts: Know what mapping, staging and meta-data in practice.
- Different Approaches: Compare warehousing, federation and virtualization.

9.2 What is it?:

The process of combining data from different sources to provide a unified and consistent view. The end goal of this is to make data usuable for analytics operations and descion-making.

9.3 Why does it matter?:

- Break down silos: Ensures data is not trapped in seperate systems.
- Consistency and quality: Improves trust worthiness of analytics.

- Supports descion-making: Executives and analysts see the "big picture".
- Operational value: Customer 360 views, supply chain monitoring.
- Cross-domain insights - For example, linking health records with genomic data to improve treatment.

9.4 Challenges in Data Integration?:

Development Issues - Inconsistent field names, missing or duplicated records, and schema mismatches.
Technical Issues - Combining structured and unstructured data, scaling for big data,
balancing preloaded and real-time.
Organgisational Issues - Data ownership, access and permissions and lock of goverance.

9.5 Tech in Data Integration:

9.5.1 Traditional ETL Tools

- Mature and enterprise grade
- Provide drag-and-drop design, scheduling and monitoring
- Often used in on-prem warehouses

9.5.2 Cloud Integration

- Severless ETL and ELT services
- Integrate with cloud storage and cloud warehouses
- Offers scalability and pay-as-you-go functionality

9.6 ETL:

Extract, transform, and load is the primary method used to integrate data from multiple sources to one central one. The Bridge function: connect raw, scattered data to usable, trusted datasets.

9.6.1 Supporting ETL

Data Mapping:

- Defines how source fields map to target schema
- Ensures consistency when merging heterogeneous data

Data Staging:

- Provides a temporary area to collect, cleanse and compare data before loading
- Helps detect changes and pre-aggregate values

9.6.2 Data Extraction

Logic method - What data is extracted?

- Full extraction
  - All records are pulled each time
  - Costly for large datasets
- Incremental Extraction
  - Pull new or changed records
  - Needs change tracking features

Physical method - How it is done?

- Online (Real-time)
  - Direct full from live systems
  - APIs
  - RTMP streams
  - Adds system load
- Offline (Batch)
  - Export files or snapshots at intervals
  - Not in real-time

9.6.3 Data Trasnformation

Most complex step, consisting of the following key tasks:

- Standardisation
- Type conversions
- De-duplication
- Date scrubbing
- Enrichment

9.6.4 Data Loading

Moving transformed data into a target repo. Load types:

- Replace or overwrite
  - Old dates discarded and replaced with the newest
  - Simple
  - Loses history
- Append (Insert)
  - New recoreds added to existing dataset
  - Useful for logs or transactional data
- Maintain historical versions
  - Store old and new data
  - Slowly changing dimensions
  - Supports trend analysis and auditing

9.7 Need Data Integration:

It is done for providing data in a specific view as requested by end users. The bigger an organisation gets, the more data and variety it uses. Increased with the need for data sharing.

9.8 Data Integration Approaches:

9.8.1 Federated DBs

- Provides virtual integration layers across multiple autonomous DBs.
- Data remains on original systems but common interfacts lets users use it as if it was a single DB.
- No need to move or duplicate data. But performance can be slower and dependent on source availability.

9.8.2 Memory-Mapped Structures

- Integration through fast, in-memory data processing.
- Data loaded into memory.
- High performance but limited by amount of memory available.

9.8.3 Data Warehousing

- A centralise repo where data from multiple source is ETL.
- High quality, consistent and analysis ready data.
- At the cost of upfront effect and ongoing maintainence.

9.9 Meta Data:

Basically just data about data. It is information that describes content, structure and context about data. Types of meta-data can include,

- Administrative
  - Information about ownership and permissions
  - Supports compliance, accountablility and secure data management
- Process
  - Describes how it was created and transformed through ETL
- Technical
  - Details about structure and storage of data
- Application
  - Records how data was used and accessed

10 Data Security and Privacy

10.1 Getting Data:

When data is sourced from multiple areas, it is possible to re-identifly anonymised data. For example using credit car information, to link with a postal address, allowing rego plate identification and knowledge about workplace and family members. This can be done through our use of everyday apps. These leave data trails everywhere.

10.2 The Fundamental Right:

The right to respect a private life, and home. Some people think this is a fundamental right that every company follows. If companies meet certain requirements then the Government gives them free reign over out data.

10.3 Data Protection:

Personal Data: Any information that indentifies or relates to an individual.
Principles: Consent, transparency and security.
UE Directive (1995): Data must only be collected and used or a specific purpose. Once completed this data should ideally be deleted.

There is no way to avoid data collection. Users mest take it upon themselves to protect their data.

10.4 Significant Risk of Big Data:

Big data poses significant risks for the protection of peronsal data:

- Large-scale traking, profiling and surveillance
- Data security and transperancy issues
- Risks of inaccuracy, bias, discrimmination and exclusion
- Lack of harmonised laws across countries

10.5 Harmonization Laws:

The first steps to the standardization of data protection standards. As personal data is collected and shared across sectorial and national boundaries, inconsistent data protection standards pose threats to privacy to individuals.

10.6 Core Security Principles:

Remember CIA:

10.6.1 Confidentiality

The neeed to keep confidential data is only available to the correct people. They also must using secure systems for storing this information to external and interal threats.

10.6.2 Integrity

- Verify data is correctly formatted
- Inputted data is accurate
- Data is following organistional rules
- Record who changed what and where

10.6.3 Availability

- Data needs to be available at all times
- Available to appropiate users
- Track who has acess and when they accessed it

10.7 Middleware Security Concerns:

A major security issue comes from a middleware attack. Anything that sits between users and access.

10.8 Security Solution Description:

Solution Description	Pros	Cons
Data Obfuscation (Masking, Scrambling)	Fake or Scrambled data set for use by design and implementation teams	Can be very expensive - good fake data can range in cost from $200,000 to $1 Million
Encryption of Data	Allows personally identifiable data to be scrambled if intrusion takes place.	Adds overhead and possible performance issues.
Database Intrusion/Extrusion Prevention	Looks for SQL Injections, Bad access commands and odd outbound data	Can eat into over head and cause performance issues - also expensive. Needs very specific criteria to set up.
Data Leak Prevention	Catches any data that is being sent out of the system	Does not protect data in the actual data warehouse.

10.9 Using Built-in Solutions:

Solution Description	Pros	Cons
Complex Passwords (require numbers and symbols) as well as frequent password changes	Makes passwords harder to guess and harder to crack	Users write them down and keep them next to computer or forget and need multiple resets
Keep Internal and External facing databases separate	Makes it very hard to hack one and then get through to the other	Reduces functionality of databases and restricts flow of internal data
Restrict Downloading	Keeps data in the database and not loose in excel, etc	Restricts reporting capabilities and off line functionality
Restrict Unwanted Connections	Again makes it harder to worm from one system to another	Makes integration more difficult and can reduce user acceptance
SAML (Security Assertion Markup Language)	SAML is the standard that is used for Single Sign On functionality	If not in use blocks the usage of single sign on

10.10 Privacy Facets:

- Anonymity: Concealing identity of an entity
- Pseudonymity: Concealing true idenity of an entity
- Fair Use: Data collection relevant to the context
- Access: Limited access to dhe data
- Control Aggregation: Restricted distribution and appropiate sentitization
- Control Lifecycle: Data operationalization management

10.11 Privacy Issues:

Using cloud services

- Entities outside of the organization now store and compute the data
- Attackers target the link between the cloud and client
- Cloud provider can now be phished
- Hard to audit data security
- Who is responsible for the when it leaks?
- Cloud can sub-contract to a third party organization

11 Machine Learning Applications

11.1 Watershed Moment:

In November 2016, Google integrated a multilingual nueral machine translation into Google translate. One of the first large-scale successful uses of deep nueral networks. This caused international interest for to adopt ML.

11.2 Evolution of Machine Learning:

ML is used everywhere within life. While most people think ML is just an algorithm, that is only one part of it. Complete ML systems also include business requirements, UI/UX, data pipelines and infrastructure which supports them.

11.3 When to use Machine Learning?:

ML is a powerful tool, but not always the best solution. It must first be considered whether it is truly necessary or cost-effective before starting a ML project.

11.4 Learn:

The most important step in ML. ML systems require large amounts of data to learn. In supervised learning, they learn from input-output pairs. This can then be used to predict output based on novel input.

11.5 Existing Data:

ML models can be deployed without prior training and use live data. This is known as continual learning. Deploying under trained models risks poor performance and bad UX.

11.6 Unseen Data:

The patterns a model learns on requires the assumption that the unseen data follow similar trends.

11.7 Where ML Works Best?:

The models work best when it's repetitive. This is becausethe patterns to be found are repeated multiple times. Also when the cost of wrong prediction is cheap. Good uses include Movies, music and clothes recommendations. Rather then Health data, Tesla's FSD.

11.8 ML vs Traditional Software:

- ML production would improve if ML experts had stronger software engineering skills. Many traditional SWE tools can support ML application development and deployment.
- In ML, testing and versioning data is critical but difficult—large datasets raise challenges: How do we version them? How do we assess if a sample is good or bad? Not all data is equal; some samples contribute more value to model performance than others.
- Model size is another hurdle. Modern ML models often have hundreds of millions or even billions of parameters, requiring gigabytes of RAM. What seems large today may look trivial in the near future—just as the computer that powered the Apollo missions had only 32 MB of RAM.